Numerical Method

OR

Numerical Analysis

Syllabus :

1. Introduction and Overview
2. Error and Approximation
3. Roots of Non Linear Equation
4. Solution of System of Linear Equation
5. Numerical Interpolation
6. Numerical Integration
7. Numerical Differentiation
8. Numerical Solution of ODE
9. Boundary Value Problem

② Error And Approximation

→ Exact & Approximate Number

→ Significant figures

→ Types of Errors

  → Inherent Error

  → Round-off Error

  → Truncation Error

  → Absolute Error

  → Relative Error

  → Percentage Error

③ Roots Of Non-Linear Equation

- → Bisection Method
- → False position or Regula Falsi Method
- → Newton Raphson Method
- → Secant Method
- → Muller's Method
- → Fixed point Method

④ Solution Of System of Linear Equation

- → Gauss Elimination Method
- → Gauss Jordan Method
- → LU decomposition Method
- → Matrix Inverse Method
- → Jacobi Iteration Method
- → Gauss Seidel Method

⑤ Numerical Interpolation

- → Operators In Interpolation
- → Newton Forward Interpolation
- → Newton Backward Interpolation
- → Gauss Forward Interpolation
- → Gauss Backward Interpolation
- → Stirling Interpolation
- → Bessel's Interpolation
- → Lagrange's Interpolation
- → Newton Divided Difference
- → Lagrange's Inverse Interpolation
- → Missing term Problem

⑥ Numerical Integration

- → Newton-Cotes formula
- → Trapezoidal Rule
- → Simpson's $\frac{1}{3}$ Rule
- → Simpson's $\frac{3}{8}$ Rule
- → Boole's Rule
- → Weddle's Rule
- → Mid-point Rule
- → Double Integration Problem
- → Romberg Integration
- → Euler Maclaurin Integration
- → Gauss Legendre (2, 3, 4, 5 points) Integration

7 Numerical Differentiation

→ Numerical differentiation Using
Newton forward & Backward Formula

→ Numerical differentiation Using Stirling Formula

→ Numerical differentiation Using
Newton divided difference Formula

→ Richardson Extrapolation

8 Numerical Solution Of Ordinary differential Equation

- → Taylor's Series Method
- → Picard Method
- → Euler's Method
- → Modified Euler's Method
- → Runge-Kutta Second Order Method
- → Runge-Kutta forth Order Method
- → Milne's predictor Corrector Method
- → Adams-Bashforth Method

9 Boundary Value Problem

→ Finite difference Method

Connect with me

Start Practicing

i._am._arfin

Arfin Parween

Arfin Parween (Subscribe here too)

Arfin Parween

Arfin Parween